Mathematical Methods · 1. 18/Oct.

1(a). ① $\underline{s}_1 + \underline{s}_2 \in S$ for all $\underline{s}_1, \underline{s}_2 \in S$.

② $\alpha \underline{s} \in S$ for all $\alpha \in F$ and $\underline{s} \in S$.

~~with~~ vector addition:

$(\underline{s}_1, \underline{s}_2) \mapsto \underline{s}_1 + \underline{s}_2 \in S$, where $\underline{s}_1, \underline{s}_2 \in V$.

Scalar multiplication.

$(\alpha, \underline{v}) \mapsto \alpha \underline{v} \in V$, where $\alpha \in F$ and $\underline{v} \in V$.

$F(S,V)$ is closed under vector addition and scalar multiplication, so ~~(b)~~ it is a vector space.

(b). ① Even functions.

let $S = [-1, 1]$, $V = \mathbb{R}$.

Define $W = \{ f: S \to \mathbb{R} \mid f(-x) = f(x) \}$.

Check: ~~$f(-x_1) + f(-x_2) = f(x_1$~~

~~$f(-x) + g(-x)$~~

$(f+g)(-x) = f(-x) + g(-x) = f(x) + g(x) = (f+g)(x)$.

and $(\alpha f)(-x) = (\alpha f)(x) = \alpha f(x)$. ~~$= \alpha f(-x)$~~.

∴ It is a vector space.

② Odd functions.

let $S = [-1, 1]$, $V = \mathbb{R}$.

Define $W = \{ f: S \to \mathbb{R} \mid f(-x) = -f(x) \}$.

check: $(f+g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f+g)(x)$

$(\alpha f)(-x) = \alpha f(-x) = -\alpha f(x)$

∴ It is a vector space.

③. A continuous function.

let $S = [-1, 1]$, $V = \mathbb{R}$.

Define $W = \{ f: [-1, 1] \to \mathbb{R} \mid f$ is ~~continuous~~ continuous $\}$.

$\alpha f(x_1) + \beta f(x_2)$ is continuous with $\alpha, \beta \in F$.

∴ It is a vector space.

(c). They all satisfy

Norm	inner product
$\sqrt{\int_{-1}^{1} f^2 dx}$.	$\int_{-1}^{1} fg\, dx$.

2(a). $RHS = \frac{1}{4}\left(\langle \underline{v}, \underline{w} \rangle - \langle \underline{v}, -\underline{w} \rangle - i\langle \underline{v}, i\underline{w} \rangle + i\langle \underline{v}, -i\underline{w} \rangle \right)$

$= \frac{1}{4}\left(\langle \underline{v}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle \right)$

$= \langle \underline{v}, \underline{w} \rangle = LHS.$

(b). For real innear product space.

$\langle \underline{v}, \underline{w} \rangle = \|\underline{v} + \underline{w}\|^2 = \|\underline{v}\|^2 + \|\underline{w}\|^2 + 2\langle \underline{v} | \underline{w} \rangle.$

2(a) $RHS = \frac{1}{4}\left[\langle \underline{v}+\underline{w}, \underline{v}+\underline{w} \rangle - \langle \underline{v}-\underline{w}, \underline{v}-\underline{w} \rangle - i\langle \underline{v}+i\underline{w}, \underline{v}+i\underline{w} \rangle + i\langle \underline{v}-i\underline{w}, \underline{v}-i\underline{w} \rangle \right)$

$= \frac{1}{4}\left(\|\underline{v}\|^2 + \|\underline{w}\|^2 + \langle \underline{v}, \underline{w} \rangle + \langle \underline{w}, \underline{v} \rangle - \|\underline{v}\|^2 - \|\underline{w}\|^2 + \langle \underline{v}, \underline{w} \rangle + \langle \underline{w}, \underline{v} \rangle - i\|\underline{v}\|^2 - i\|\underline{w}\|^2 + \langle \underline{v}, \underline{w} \rangle - \langle \underline{w}, \underline{v} \rangle + i\|\underline{v}\|^2 + i\|\underline{w}\|^2 + \langle \underline{v}, \underline{w} \rangle - \langle \underline{w}, \underline{v} \rangle \right)$

$= \frac{1}{4}(4\langle \underline{v}, \underline{w} \rangle)$

$= \langle \underline{v}, \underline{w} \rangle$

$= LHS.$

(b). For real inner product space.

~~$\langle \underline{v}, \underline{w} \rangle =$~~

$\|\underline{v}+\underline{w}\|^2 = \|\underline{v}\|^2 + \|\underline{w}\|^2 + 2\langle \underline{v}, \underline{w} \rangle$.

$\|\underline{v}-\underline{w}\|^2 = \|\underline{v}\|^2 + \|\underline{w}\|^2 - 2\langle \underline{v}, \underline{w} \rangle$.

$\Rightarrow \langle \underline{v}, \underline{w} \rangle = \frac{1}{4}(\|\underline{v}+\underline{w}\|^2 - \|\underline{v}-\underline{w}\|^2)$.

(c). ~~$\langle \underline{v}, T\underline{w} \rangle = \frac{1}{4}(\langle \underline{v}+\underline{w}, T(\underline{v}+\underline{w})$~~ T is linear: $T(\underline{v}+\underline{w}) = T(\underline{v}) + T(\underline{w})$.

RHS $= \frac{1}{4}(\langle \underline{v}, T(\underline{v}) \rangle + \langle \underline{v}, T(\underline{w}) + \langle \underline{w}, T(\underline{v}) \rangle + \langle \underline{w}, T(\underline{w}) \rangle$

$- \langle \underline{v}, T\underline{v} \rangle + \langle \underline{v}, T\underline{w} \rangle + \langle \underline{w}, T\underline{v} \rangle - \langle \underline{w}, T\underline{w} \rangle$

$- i\langle \underline{v}, T\underline{v} \rangle + \langle \underline{v}, T\underline{w} \rangle + \langle \underline{w}, T\underline{v} \rangle - \langle \underline{w}, T(i\underline{w}) \rangle$

$+ i\langle \underline{v}, T\underline{v} \rangle - i\langle \underline{v}, T(i\underline{w}) \rangle - \langle \underline{w}, T\underline{v} \rangle + \langle \underline{w}, T(i\underline{w}) \rangle$

$= \frac{1}{4} \cdot 4\langle \underline{v}, T\underline{w} \rangle$

$= \langle \underline{v}, T\underline{w} \rangle$. = LHS

(d). from (c).

$\langle \underline{v}, T\underline{w} \rangle = \frac{1}{4}(\langle \underline{v}+\underline{w}, T(\underline{v}+\underline{w}) \rangle - \langle \underline{v}-\underline{w}, T(\underline{v}-\underline{w}) \rangle$

$- i\langle \underline{v}+i\underline{w}, T(\underline{v}+i\underline{w}) \rangle + i\langle \underline{v}-i\underline{w}, T(\underline{v}-i\underline{w}) \rangle)$

$= \frac{1}{4}(4\langle \underline{v}, T\underline{v} \rangle - \langle 0, T(0) \rangle$

$- i\langle \underline{v}, T\underline{v} \rangle + \langle \underline{v}, T\underline{v} \rangle - \langle \underline{v}, T\underline{v} \rangle - i\langle \underline{v}, T\underline{v} \rangle$

$+ i\langle \underline{v}, T\underline{v} \rangle + \langle \underline{v}, T\underline{v} \rangle - \langle \underline{v}, T\underline{v} \rangle + i\langle \underline{v}, T\underline{v} \rangle)$

$=$

Choose $\underline{v} = T\underline{w}$ as ~~This~~ the polarisation identity holds for every $\underline{v}, \underline{w} \in V$.

$\therefore \langle T\underline{w}, T\underline{w} \rangle = 0$

$\|T\underline{w}\|^2 = 0$.

$T\underline{w} = 0$

$T = 0$ as $\underline{w} \neq 0$.

(e). ~~$\langle \underline{v}, T\underline{v} \rangle$~~ for $T = T^+$.

① $\langle \underline{v}, T\underline{v} \rangle = \langle T^+\underline{v}, \underline{v} \rangle = \langle T\underline{v}, \underline{v} \rangle$.

And $\langle \underline{v}, T\underline{v} \rangle = \langle T\underline{v}, \underline{v} \rangle^*$.

$\therefore \langle T\underline{v}, \underline{v} \rangle^* = \langle T\underline{v}, \underline{v} \rangle$.

$\therefore \langle \underline{v}, T\underline{v} \rangle \in \mathbb{R}$ for all $\underline{v} \in V$.

② for $\langle \underline{v}, T\underline{v} \rangle \in \mathbb{R}$.

$\langle \underline{v}, T\underline{v} \rangle^* = \langle \underline{v}, T\underline{v} \rangle$ ~~$= \langle T\underline{v}, \underline{v} \rangle$~~.

and $\langle \underline{v}, T\underline{v} \rangle^* = \langle T\underline{v}, \underline{v} \rangle$.

$\therefore \langle \underline{v}, T\underline{v} \rangle = \langle T\underline{v}, \underline{v} \rangle = \langle \underline{v}, T^+\underline{v} \rangle$.

which only holds true for $T = T^+$.

3(a). $\|\underline{v}+\underline{w}\|^2 + \|\underline{v}-\underline{w}\|^2 = \langle \underline{v}+\underline{w}, \underline{v}+\underline{w}\rangle + \langle \underline{v}-\underline{w}, \underline{v}-\underline{w}\rangle$

$= \|\underline{v}\|^2 + \|\underline{w}\|^2 + \langle \underline{v}, \underline{w}\rangle + \langle \underline{w}, \underline{v}\rangle$

$+ \|\underline{v}\|^2 + \|\underline{w}\|^2 - \langle \underline{v}, \underline{w}\rangle - \langle \underline{w}, \underline{v}\rangle$

$= 2(\|\underline{v}\|^2 + \|\underline{w}\|^2)$.

(b). ①. As $1<p<\infty$.

$|x_i|^p > 0$ for any $i \in \mathbb{N}$

$\therefore \|(x_i)\| := \left(\sum_{i=1}^{\infty} |x_i|^p\right)^{1/p} > 0$. for all $x_i \in \mathbb{C}$.

②. $\|\lambda(x_i)\| := \left(\sum_{i=1}^{\infty} \lambda^p |x_i|^p\right)^{1/p} = |\lambda| \left(\sum_{i=1}^{\infty} (x_i)^p\right)^{1/p} = |\lambda| \|(x_i)\|$.

③. $\|(x_i)\| + \|(y_i)\| := \left(\sum_{i=1}^{\infty} |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^{\infty} |y_i|^p\right)^{1/p}$

$\|(x_i)+(y_i)\| := \left(\sum_{i=1}^{\infty} |x_i+y_i|^p\right)^{1/p}$

For every $i \in \mathbb{N}$

$\|x_i+y_i\|^2 = \|x_i\|^2 + \|y_i\|^2 + 2\,\mathrm{Re}\langle x_i | y_i\rangle$

$\leq \|x_i\|^2 + \|y_i\|^2 + 2\langle x_i | y_i\rangle$.

using Schwarz's inequality.

$|\langle \underline{a} | \underline{b}\rangle| \leq \|\underline{a}\| \|\underline{b}\|$.

$\Rightarrow \|x_i+y_i\|^2 \leq \|x_i\|^2 + \|y_i\|^2 + 2\|x_i\|\|y_i\| = (\|x_i\| + \|y_i\|)^2$

$\therefore |x_i+y_i| \geq \|x_i+y_i\|$

for all $x_i, y_i \in \mathbb{C}$.

$\therefore$ the expression defines a norm on l^p.

(c). $\|\underline{v}+\underline{w}\|^2 = \left(\sum_{i=1}^{\infty} |v+w|^p\right)^{2/p} = (1^p + 1^p + 0\cdots)^{2/p}$

$\|\underline{v}-\underline{w}\|^2 = \left(\sum_{i=1}^{\infty} |v-w|^p\right)^{2/p} = (1^p + (-1)^p + 0\cdots)^{2/p}$

$\|\underline{v}+\underline{w}\|^2 + \|\underline{v}-\underline{w}\|^2 = 2(\|\underline{v}\|^2 + \|\underline{w}\|^2) = 4$. from (a).

So $2(1^p + 1^p + 0 + \cdots)^{2/p} = 4$.

$(1^p + 1^p + 0)^{2/p} = 2$.

which is true for only $p=2$.

$\therefore$ The above norm on l^p is not associated to an inner product for $p \neq 2$.

4(a). ① $\phi_0 = 1$.

$\langle \phi_0 | \phi_0 \rangle^{\frac{1}{2}} = \int_{-\infty}^{\infty} e^{-x^2} dx = \pi^{1/4}$

~~$\phi_1 = y_1 =$~~

$\hat{\phi}_0 = \frac{\phi_0}{\pi^{1/4}} = \pi^{-\frac{1}{4}}$.

② $\phi_1 = y_1 - \hat{\phi}_0 \langle \hat{\phi}_0 | y_1 \rangle = x - \pi^{-\frac{1}{4}} \langle \pi^{-\frac{1}{4}} | x \rangle$

~~$\pm$~~ $\langle \pi^{-\frac{1}{4}} | x \rangle = \int_{-\infty}^{\infty} e^{-x^2} x \pi^{-\frac{1}{4}} dx$

$= \left[-\frac{1}{2\pi^{\frac{1}{4}}} e^{-x^2} \right]_{-\infty}^{\infty}$

$= 0$.

$\therefore \phi_1 = x$.

$\|\phi_1\|^2 = \langle \phi_1, \phi_1 \rangle = \int_{-\infty}^{\infty} x^2 e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

$\Rightarrow \|\phi_1\| = \frac{\pi^{1/4}}{\sqrt{2}}$

$\hat{\phi}_1 = \frac{\phi_1}{\|\phi_1\|} = \frac{\sqrt{2} x}{\pi^{1/4}}$

③ ~~$\hat{\phi}_2 =$~~ $\phi_2 = $ ~~0~~ $y_2 - \hat{\phi}_1 \langle \hat{\phi}_1 | y_2 \rangle - \hat{\phi}_0 \langle \hat{\phi}_0 | y_2 \rangle$

$= x^2 - \frac{\sqrt{2}}{\pi^{1/4}} x \langle \frac{\sqrt{2}}{\pi^{1/4}} x | x^2 \rangle - \pi^{-\frac{1}{4}} \langle \pi^{-\frac{1}{4}} | x^2 \rangle$

④

$\langle \frac{\sqrt{2}}{\pi^{1/4}} x | x^2 \rangle = \int_{-\infty}^{\infty} e^{-x^2} x^3 \frac{\sqrt{2}}{\pi^{1/4}} dx = 0$.

which is an odd function.

$\langle \pi^{-\frac{1}{4}} | x^2 \rangle = \int_{-\infty}^{\infty} e^{-x^2} \pi^{-\frac{1}{4}} x^2 dx = \frac{\pi^{\frac{1}{4}}}{2}$

$\Rightarrow \phi_2 = x^2 - \frac{\sqrt{2}}{\pi^{1/4}} x \cdot 0 - \pi^{-\frac{1}{4}} \cdot \frac{\pi^{\frac{1}{4}}}{2} = x^2 - \frac{1}{2}$.

$\|\phi_2\|^2 = \langle \phi_2, \phi_2 \rangle = \int_{-\infty}^{\infty} e^{-x^2} (x^2 - \frac{1}{2})^2 dx$

$= \int_{-\infty}^{\infty} (e^{-x^2} x^4 - e^{-x^2} x^2 + \frac{1}{4} e^{-x^2}) dx$

$= \frac{3}{4}\sqrt{\pi} - \frac{1}{2}\sqrt{\pi} + \frac{1}{4}\sqrt{\pi} = \frac{\sqrt{\pi}}{2}$

$\hat{\phi}_2 = \frac{\phi_2}{\|\phi_2\|} = \frac{x^2 - \frac{1}{2}}{\pi^{\frac{1}{4}}} \sqrt{2}$.

Final Answer

④ $\hat{\phi}_0 = \pi^{-\frac{1}{4}}$

$\hat{\phi}_1 = \frac{\sqrt{2}}{\pi^{1/4}} x$

$\hat{\phi}_2 = \frac{\sqrt{2}}{\pi^{1/4}} (x^2 - \frac{1}{2})$.

(b). $q(x) = a_0 + a_1 x + a_2 x^2$

$q(x) = c_0\hat{\phi}_0 + c_1\hat{\phi}_1 + c_2\hat{\phi}_2$

$= c_0\pi^{-\frac{1}{4}} + c_1\sqrt{2}\pi^{-\frac{1}{4}}x + c_2\sqrt{2}\pi^{-\frac{1}{4}}(x^2 - \frac{1}{2})$

$= c_0\pi^{-\frac{1}{4}} - \frac{1}{\sqrt{2}}\pi^{-\frac{1}{4}}c_2 + c_1\sqrt{2}\pi^{-\frac{1}{4}}x + c_2\sqrt{2}\pi^{-\frac{1}{4}}x^2.$

Equate the components.

$$\begin{cases} \sqrt{2}\pi^{-\frac{1}{4}} c_2 = a_2 \\ \sqrt{2}\pi^{-\frac{1}{4}} c_1 = a_1 \\ c_0\pi^{-\frac{1}{4}} - \frac{1}{\sqrt{2}}\pi^{-\frac{1}{4}} c_2 = a_0. \end{cases}$$

$$\Rightarrow \begin{cases} c_2 = \frac{\pi^{\frac{1}{4}}}{\sqrt{2}} a_2 \\ c_1 = \frac{\pi^{\frac{1}{4}}}{\sqrt{2}} a_1 \\ c_0 = (a_0 + \frac{a_2}{2})\pi^{\frac{1}{4}}. \end{cases}$$

$$q(x) = (a_0 + \frac{a_2}{2})\pi^{\frac{1}{4}}\hat{\phi}_0 + \frac{\pi^{\frac{1}{4}}}{\sqrt{2}} a_1 \hat{\phi}_1 + \frac{\pi^{\frac{1}{4}}}{\sqrt{2}} a_2 \hat{\phi}_2.$$

(c). $\|q\|^2 = \langle q | q \rangle = \langle \sum_i c_i\hat{\phi}_i, \sum_j c_j\hat{\phi}_j \rangle = c_0^2 + c_1^2 + c_2^2$

$= (a_0 + \frac{a_2}{2})^2\pi^{\frac{1}{2}} + \frac{\pi^{\frac{1}{2}}}{2} a_1^2 + \frac{\pi^{\frac{1}{2}}}{2} a_2^2.$

(d). $\int_{-\infty}^{\infty} \langle f', g' \rangle^* (T\langle f, g \rangle)\, dx = \int_{-\infty}^{\infty} \langle f', g' \rangle^* (\frac{1}{w}\frac{d}{dx} w \frac{d}{dx} \langle f, g \rangle)\, dx$

$= [\langle f', g' \rangle^* \frac{1}{w}\frac{d}{dx} w \langle f, g \rangle]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{d}{dx}\langle f', g' \rangle^* \frac{1}{w}\frac{d}{dx} w \langle f, g \rangle\, dx$

~~$= 0 - \int_{-\infty}^{\infty} (\frac{1}{w}\frac{d}{dx} w \frac{d}{dx} \langle f', g' \rangle^*) \langle f, g \rangle\, dx$~~

$= [-2x \langle f', g' \rangle^* \langle f, g \rangle]_{-\infty}^{\infty} -$

?

5(a). $|\xi_i\rangle\langle\xi_i| = id.$

$\therefore RHS = \Sigma_i \langle v|\hat{I}|w\rangle = \langle v|w\rangle = LHS.$

(b). $\langle \xi_i' | \xi_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{for } i=j \\ 0 & \text{for } i\neq j. \end{cases}$ $\therefore P_{ii} = 1$

$\therefore$ P is unitary.

(c) RHS =

$P_{ij} = \langle \xi_i | \hat{P} | \xi_j \rangle$

$\Sigma_j (P^\dagger)_{ij} P_{jk} =$

$\Sigma_j (P^\dagger)_{ij} P_{jk} = \Sigma_j \langle \xi_i | \hat{P}^\dagger | \xi_j \rangle \langle \xi_j | \hat{P} | \xi_k \rangle = \langle \xi_i | \hat{P}^\dagger \hat{P} | \xi_k \rangle$

$= id.$

$P_{ij} = \langle \xi_i' | \xi_j \rangle.$

$P^\dagger_{ij} = \langle \xi_i' | \xi_j \rangle^\dagger = \langle \xi_j | \xi_i' \rangle^{-1}$

(b) $PP^\dagger = P_{ij} P^\dagger_{ij} = \langle \xi_i' | \xi_j \rangle \langle \xi_j | \xi_i' \rangle^{-1} = \langle \xi_i' | \xi_i' \rangle^{-1} = id.$ ($= id.$)

$\therefore$ P is unitary.

(c). $RHS = P_{ij} T_{jk} P^\dagger_{kl} = \langle \xi_i' | \xi_j \rangle \langle \xi_j | \hat{T} | \xi_k \rangle \langle \xi_k' | \xi_l \rangle^\dagger$ ($= id.$) ($= id.$ from (b))

$= \langle \xi_i' | \hat{T} | \xi_k \rangle [\langle \xi_l | \xi_k' \rangle]^{-1}.$ ($= id$)

$= \langle \xi_i' | \hat{T} | \xi_k' \rangle$

$= T'_{ik}$

$= T' = LHS.$

6(a). As T is purely imaginary, $T^* = -T$.

$$R = \mathbb{I}_3 + iT + O(T^2)$$
$$R^T = \mathbb{I}_3 + iT^{*T} + O(T^2)$$
$$R^T R = \mathbb{I}_3 + i(T + T^{*T}) + O(T^2)$$

for R satisfy $R^T R = \mathbb{I}_3$ and $\det(R) = 1$.

$$i(T + T^{*T}) = 0.$$
$$T = -T^T$$

$\therefore$ T is anti-symmetric.

(b). $[\tilde{T}_i, \tilde{T}_j]_{lm} = (\tilde{T}_i)_{ln}(\tilde{T}_j)_{nm} - (\tilde{T}_j)_{ln}(\tilde{T}_i)_{nm}$

$$= (-i\varepsilon_{iln})(-i\varepsilon_{jnm}) - (-i\varepsilon_{jln})(-i\varepsilon_{inm})$$
$$= -\varepsilon_{nil}\varepsilon_{nmj} + \varepsilon_{njl}\varepsilon_{nmi}$$
$$= -\delta_{im}\delta_{lj} + \delta_{ij}\delta_{lm} + \delta_{jm}\delta_{li} - \delta_{ji}\delta_{lm}$$
$$= \delta_{jm}\delta_{li} - \delta_{im}\delta_{lj}$$
$$= \varepsilon_{qil}\varepsilon_{qmj}$$

$$= \delta_{jm}\delta_{li} - \delta_{im}\delta_{lj}$$

$$[i\varepsilon_{ijk}\tilde{T}_k]_{lm} = \varepsilon_{ijk}\varepsilon_{klm} = \varepsilon_{kij}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$$

$\therefore [\tilde{T}_i, \tilde{T}_j]_{lm} = [i\varepsilon_{ijk}\tilde{T}_k]_{lm}$

$$[\tilde{T}_i, \tilde{T}_j] = i\varepsilon_{ijk}\tilde{T}_k.$$

$\begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}\begin{bmatrix} 0 & -b \\ -a & 0 \end{bmatrix}$.

(c) $U^+ = \mathbb{I}_2 + iS^+$

$$U^+U = (\mathbb{I}_2 + iS)(\mathbb{I}_2 + iS^+) = \mathbb{I}_2 + SS^+ + i(S + S^+) = \mathbb{I}_2.$$

$\therefore S + S^+ = 0.$ $\Rightarrow S$ is Hermitian.

$SS^+ = 0.$ $\Rightarrow \mathrm{tr}(S) = 0.$

(d). ① $\underline{\sigma_1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. $\underline{\sigma_2} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. $\underline{\sigma_3} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

$\underline{\tau_1} = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\underline{\tau_2} = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. $\underline{\tau_3} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

$\underline{\tau} = \sum_{i=1}^{3} \alpha_i \underline{\tau_i}$

As $\underline{\tau} = \underline{0}$ only if $^{all}\alpha_i = 0$. ; And Pauli matrices are hermitian, traceless

$\therefore \tau_i$ form a basis of the space of 2×2 hermitian traceless matrices.

$$[\tau_i, \tau_j]_{lm} = (\tau_i \tau_j)_{lm} - (\tau_j \tau_i)_{lm}$$
$$= (\tau_i)_{lk}(\tau_j)_{km} - (\tau_j)_{lk}(\tau_i)_{km}$$

② $\varepsilon_{ijk} = \begin{cases} 1 & \text{if } (i,j,k) \text{ is an even permutation of } (1,2,3). \\ -1 & \text{if } (i,j,k) \text{ is an odd permutation of } (1,2,3) \\ 0 & \text{otherwise (two indices are equal)}. \end{cases}$

To prove. $[\tau_1, \tau_2] = i\varepsilon_{123}\tau_3 = i\tau_3$

$[\tau_2, \tau_3] = i\varepsilon_{231}\tau_1 = i\tau_1$

$[\tau_3, \tau_1] = i\varepsilon_{312}\tau_2 = i\tau_2$.

$$[\tau_1, \tau_2] = \frac{1}{4}\left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right]$$
$$= \frac{1}{4}\left[\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}\right] = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\tau_3$$

$$[\tau_2, \tau_3] = \frac{1}{4}\left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\right]$$
$$= \frac{1}{4}\left[\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}\right] = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\tau_1$$

$$[\tau_3, \tau_1] = \frac{1}{4}\left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right]$$
$$= \frac{1}{4}\left[\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\right] = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\tau_2$$

$\therefore [\tau_i, \tau_j] = i\varepsilon_{ijk}\tau_k$.

(e). ① $r(\tilde{T}_i) := \tau_i$ is ~~bj~~ bijective.

② $r([\tilde{T}_i, \tilde{T}_j]) = r(i\varepsilon_{ijk}\tilde{T}_k) = i\varepsilon_{ijk}\, r(\tilde{T}_k) = i\varepsilon_{ijk}\tau_k = [\tau_i, \tau_j]$.

$\therefore r([\tilde{T}_i, \tilde{T}_j]) = [r(\tilde{T}_i), r(\tilde{T}_j)]$.

7(a). ~~$|x^{k_1} - x^{k_2}| = |x_1^k - x_2^k| =$~~

① $\langle f, f \rangle = \int_0^a f(x)^2\, dx$.

$\|x^k - 0\|$

$\|x^k\| = \sqrt{\int_0^a x^{2k}\, dx} = \sqrt{\left[\frac{x^{2k+1}}{2k+1}\right]_0^a} = \sqrt{\frac{a^{2k+1}}{2k+1}}$.

As $0 < a < 1$. for $k \to \infty$.

$\lim_{k\to\infty} \|x^k\| \to 0$.

$\therefore x^k \to 0$.

$\therefore \|x^k - 0\| \to 0$.

② $\|x^m - x^n\| \le \|x^m\| + \|x^n\|$ for $m, n > N$.

As $\|x^k\| \to 0$.

$\therefore$ for any $\epsilon > 0$, there exist N for when $k > N$.

$\|x^k\| < \frac{\epsilon}{2}$.

$\therefore \|x^m - x^n\| \le \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.

$\therefore (x^k)_{k=1}^{\infty}$ is a Cauchy sequence.

(b). ~~$\|S_m - S_n\| \leq \|S_m\| + \|S_n\| \equiv$~~ for $m > n$.

$S_m - S_n = x^m + x^{m-1} + \cdots + x^{n+1}$

$= \dfrac{x^{n+1}(1 - x^{m-n})}{1 - x}$

As $0 \leq x \leq a < 1$.

$\therefore 0 \leq 1 - x^{m-n} \leq 1$.

and $1 - x \geq 1 - a > 0$.

$\therefore S_m - S_n \leq \dfrac{a^{n+1}}{1-a}$

$\|S_m - S_n\| = \sqrt{\int_0^a |S_m(x) - S_n(x)|^2 dx} \leq \sqrt{\int_0^a \dfrac{a^{2n+2}}{(1-a)^2} dx}$

$= \sqrt{\dfrac{a^{2n+3}}{(1-a)^2}}$

$\therefore \|S_m - S_n\| \leq \dfrac{\sqrt{a}\, a^{n+1}}{1-a}$.

as $a^{n+1} \to 0$.

$\|S_m - S_n\| < \varepsilon$.

$\therefore (S_k)_{k=0}^{\infty}$ is a Cauchy sequence.

(c). $S_m = 1 + x + x^2 + \cdots + x^m = \dfrac{1 - x^{m+1}}{1 - x} \to \dfrac{1}{1-x}$.

which is not a polynomial.

$\therefore S_m \notin V$. the limit does not belong to V.

$\therefore V$ is not complete and hence not a Hilbert Space.